Math 150A Section 8 and 16
Fall 2025
Final Exam Practice
May 6, 2025
Time Limit: 1 hour and 50 Minutes

Name (Print): ____________________

Student ID: ____________________

This exam contains 9 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam, unless otherwise stated. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- Unless otherwise stated, all angles are in **radians**

Problem	Points	Score
1	20	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
Total:	90	

Do not write in the table to the right.

1. (20 points) **TRUE or FALSE!** If the statement is true, write TRUE (all caps, all spelled out). Otherwise write FALSE (all caps, all spelled out).

   (a) $f(x) = |x|$ has a critical point T

   (b) $\cos(x)$ is concave down at $x = 0$ T

   (c) if $f'(2)$ exists, then $f(x)$ is continuous at $x = 2$ T

   (d) if $f'(1/2) = 2$ then the derivative of $f(\sin(x))$ at $x = \pi/6$ is $\sqrt{3}$ T

   (e) the functions $f(x) = x$ and $g(x) = x^2/x$ are the same F

   (f) $\int_0^1 f(x)^2 dx = \left(\int_0^1 f(x)dx\right)^2$ F

   (g) $\frac{d}{dx}\sin(2x) = \cos(2x)$ F

   (h) if $f'(x) = g'(x)$ then $f(x) = g(x)$ F

   (i) the function $x^{100} - 9x^2 + 1$ has a root in the interval $[0, 2]$ T

   (j) a function can never cross its horizontal asymptote F

2. (10 points) Calculate the following. Carefully show you work.

(a) $\lim_{x\to 3} \frac{\sqrt{x+6}-\sqrt{3x}}{x-3}$

$$\begin{aligned}
\lim_{x\to 3} \frac{\sqrt{x+6}-\sqrt{3x}}{x-3} &= \lim_{x\to 3} \frac{\sqrt{x+6}-\sqrt{3x}}{x-3} \frac{\sqrt{x+6}+\sqrt{3x}}{\sqrt{x+6}+\sqrt{3x}} \\
&= \lim_{x\to 3} \frac{6-2x}{(x-3)(\sqrt{x+6}+\sqrt{3x})} \\
&= \lim_{x\to 3} \frac{-2}{\sqrt{x+6}+\sqrt{3x}} = -\frac{1}{3}
\end{aligned}$$

(b) $\lim_{x\to 1} \frac{\frac{x+3}{x}-\frac{4}{x}}{x-1}$

$$\begin{aligned}
\lim_{x\to 1} \frac{\frac{x+3}{x}-\frac{4}{x}}{x-1} &= \lim_{x\to 1} \frac{\frac{x-1}{x}x}{x-1} \\
&= \lim_{x\to 1} \frac{1}{x} = 1
\end{aligned}$$

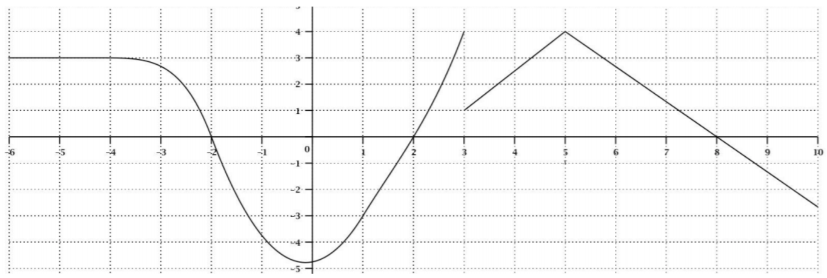
(c) $\lim_{x\to 0+} x\ln(x)$

$$\begin{aligned}
\lim_{x\to 0+} x\ln(x) &= \lim_{x\to 0+} \frac{\ln(x)}{x^{-1}} \\
&= \lim_{x\to 0+} \frac{1/x}{-x^{-2}} \\
&= \lim_{x\to 0+} -x = 0
\end{aligned}$$

(d) $\lim_{x\to 1+} x^{1/(1-x)}$

$$\begin{aligned}
\lim_{x\to 1+} x^{1/(1-x)} &= e^{\lim_{x\to 1+} \ln(x^{1/(1-x)})} \\
&= e^{\lim_{x\to 1+} \frac{\ln(x)}{1-x}} \\
&= e^{\lim_{x\to 1+} \frac{1/x}{-1}} = e^{\frac{1/1}{-1}} = e^{-1}
\end{aligned}$$

3. (10 points) The following is a graph of the **derivative** $f'(x)$ of a function $f(x)$.



(a) List all intervals where $f(x)$ is decreasing $(-2, 2)$ and $(8, \infty)$, where the derivative is positive

(b) List all intervals where $f(x)$ is concave up $(0, 5)$, where the derivative is increasing

(c) List all critical points of $f(x)$ and classify them as local max, local min or neither

- -2 is a local max by 1st derivative test
- 2 is a local min by 1st derivative test
- 3 is neither by 1st derivative test
- 8 is a local max by 1st derivative test

(d) What is $\lim_{x\to 8} \frac{f'(x)}{x-8}$? By L'Hospital:

$$\lim_{x\to 8} \frac{f'(x)}{x-8} = \lim_{x\to 8} \frac{f''(x)}{1} = f''(8)$$

This is the slope of the derivative at $x = 8$, which is $-4/3$.

(e) If $f(0) = 4$, is $f(8)$ positive or negative? We calculate

$$f(8) - f(4) = \int_4^8 f'(x)dx = \frac{1}{2}(2.5 + 4) * 1 + \frac{1}{2} * 4 * 3 = 9.25$$

and therefore

$$f(8) = f(4) + 9.25 = 4 + 9.25 = 13.25 > 0$$

4. (10 points)

   1. Find all critical points of the function $f(x) = xe^{-x}$ Taking the derivative and setting it equal to zero, we get
   $$e^{-x} - xe^{-x} = 0.$$
   Multiplying by e^x, this gives
   $$1 - x = 0.$$
   This gives us one critical point at $x = 0$.

   2. If $f(2) = 0.75$ and $f'(2) = -0.25$, use the linearization of $f(x)$ at $x = 2$ to estimate $f(1.9)$. The linearization formula is
   $$L(x) = f(a) + f'(a)(x - a) = 0.75 - 0.25(x - 2) = -0.25x + 1.25.$$
   Therefore
   $$f(1.9) \approx L(1.9) = -0.25 * 1.9 + 1.25 = 0.775$$

5. (10 points) Find the derivative of each of the following functions.

(a) $\sin^{-1}(x\cos(x))$

$$\frac{1}{\sqrt{1-x^2\cos^2(x)}}(\cos(x)-x\sin(x))$$

(b) $\frac{xe^x}{\sin(x)+\cos(x)}$

$$\frac{(e^x+xe^x)(\sin(x)+\cos(x))-xe^x(\cos(x)-\sin(x))}{(\sin(x)+\cos(x))^2}$$

(c) $\sin(x)^{\cos(x)}$ We let $y=\sin(x)^{\cos(x)}$, so that $\ln y=\cos(x)\ln\sin(x)$. Therefore

$$\frac{1}{y}y' = -\sin(x)\ln\sin(x)+\frac{\cos^2(x)}{\sin(x)}$$

so that

$$y' = y\left(-\sin(x)\ln\sin(x)+\frac{\cos^2(x)}{\sin(x)}\right)$$

or rather

$$y' = \sin(x)^{\cos(x)}\left(-\sin(x)\ln\sin(x)+\frac{\cos^2(x)}{\sin(x)}\right)$$

6. (10 points) Consider an isosceles triangle with base width of 40 and height of 60. What is the largest possible area of a rectangle inscribed in this triangle?

   Let x be the base of the rectangle and y the height. Then by similar triangles

   $$\frac{60-y}{x} = \frac{60}{40} = \frac{3}{2}.$$

   Therefore $y = 60 - 3x/2$ so that the area of the rectangle is

   $$A = xy = 60x - 3x^2/2.$$

   To optimize this function, we take the derivative and set it equal to zero, getting

   $$A' = 60 - 3x = 0.$$

   This gives $x = 20$ and $y = 30$, so that the maximum area is $A = 600$.

7. (10 points) Find the area of the region bounded by the graphs $y = x^2 - x - 1$ and $y = 1 - x^2$ in the x, y–plane.

   The two curves intersect where

$$x^2 - x - 1 = 1 - x^2,$$

   which is where

$$2x^2 - x - 2 = 0$$

   The solutions of this are

$$x = \frac{1 \pm \sqrt{17}}{4}.$$

   Therefore the area is

$$A = \int_{(1-\sqrt{17})/4}^{(1+\sqrt{17})/4} (1 - x^2) - (x^2 - x - 1)dx = \int_{(1-\sqrt{17})/4}^{(1+\sqrt{17})/4} 2 + x - 2x^2 dx = \frac{17\sqrt{17}}{24}$$

8. (10 points) Compute each of the following integrals. Show all work. Lack of work results in no credit!!

(a) $\int_1^8 \frac{3+\sqrt{t}}{t} dt$

$$\begin{aligned}\int_1^8 \frac{3+\sqrt{t}}{t} dt &= \int_1^8 3t^{-1} + t^{-1/2} dt \\ &= 3\ln t + 2\sqrt{t}|_1^8 = -2 + 4\sqrt{2} + \ln(512)\end{aligned}$$

(b) $\int \frac{2x}{\sqrt{x^2+1}} dx$ Using the u-sub $u = x^2+1$,

$$\int \frac{2x}{\sqrt{x^2+1}} dx = \int_1 \sqrt{u} du = 2\sqrt{u} + C = 2\sqrt{x^2+1} + C$$

(c) $\int \frac{3z+7}{z^2+2z+10} dz$

$$\int \frac{3z+7}{z^2+2z+10} dz = \int \frac{3(z+1)+4}{(z+1)^2+9} dz.$$

Now do a u-sub $3u = z+1$, to get

$$\begin{aligned}\int \frac{3z+7}{z^2+2z+10} dz &= \int \frac{9u+4}{9u^2+9} 3du \\ &= \int \frac{3u+4/3}{u^2+1} du \\ &= 3\int \frac{u}{u^2+1} du + \frac{4}{3}\int \frac{1}{u^2+1} du \\ &= \frac{3}{2}\ln(u^2+1) + \frac{4}{3}\tan^{-1}(u) + C \\ &= \frac{3}{2}\ln((z+1)^2/9+1) + \frac{4}{3}\tan^{-1}((z+1)/3) + C\end{aligned}$$

(d) $\int_0^3 (\sqrt{9-x^2} + x + 1) dx$

$$\begin{aligned}\int_0^3 (\sqrt{9-x^2} + x + 1) dx &= \int_0^3 \sqrt{9-x^2} dx + \int_0^3 x + 1 dx \\ &= \frac{1}{4}\pi 3^2 + \frac{9}{2} + 3\end{aligned}$$